

# Teams Bridge for IsPilot - Microsoft 365 Agents SDK

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**Status:**  New architecture using Microsoft 365 Agents SDK

**Target Deployment:** Cloud Run service `teams-ispilot-bridge`

**Complete Setup Guide:** See [../docs/TEAMS\\_SDK\\_INTEGRATION.md](https://github.com/microsoft/teams-ispilot-bridge/blob/main/docs/TEAMS_SDK_INTEGRATION.md)

This service bridges Microsoft Teams with the IsPilot API using the **official Microsoft 365 Agents SDK** (Python). It provides native Teams integration, proper JWT validation, and seamless authentication to the backend `ispilot-api`.

## Quick Links

- **Full Setup:** [../docs/TEAMS\\_SDK\\_INTEGRATION.md](https://github.com/microsoft/teams-ispilot-bridge/blob/main/docs/TEAMS_SDK_INTEGRATION.md) - Complete 5-phase implementation guide
- **Architecture Rationale:** [../docs/ARCHITECTURE\\_RESET\\_TEAMS\\_SDK.md](https://github.com/microsoft/teams-ispilot-bridge/blob/main/docs/ARCHITECTURE_RESET_TEAMS_SDK.md) - Why Teams SDK + Teams Bridge
- **API Documentation:** [../ispilot-api/README.md](https://github.com/microsoft/ispilot-api/blob/main/README.md) - `ispilot-api` endpoint reference
- **Work Checkpoint:** [../WORK\\_CHECKPOINT.md](https://github.com/microsoft/teams-ispilot-bridge/blob/main/docs/WORK_CHECKPOINT.md) - Current status and remaining tasks

## Architecture

```
Teams User
  ↓
Azure Bot Service (validates Teams JWT)
  ↓
Teams Bridge (Cloud Run - Public)
├─ MS 365 Agents SDK
├─ Validates Azure JWT
├─ Generates Google identity token (sa-tot-osa)
├─ Routes to ispilot-api
├─ Manages session context
  ↓
IsPilot API (Cloud Run - Private)
├─ Validates identity token
├─ Routes to Vertex AI Reasoning Engine
  ↓
Response → Back to Teams
```

## Prerequisites

Google Cloud Setup (Already in Place ☒)

- **Project:** `corp-stro-salesinventory-prod`
- **Region:** `us-central1`
- **Service Account:** `sa-tot-osa@corp-stro-salesinventory-prod.iam.gserviceaccount.com`
- **ispilot-api:** <https://ispilot-api-46y2f3tyja-uc.a.run.app>

## Azure / Microsoft 365 Setup (In Progress ☒ Partially Done)

**Who:** Azure/Teams admin (José Arturo)

### Completed:

- ☒ Azure AD App Registration: `ftc-ispilot-corp-prod`
- ☒ Microsoft App ID: `53ead9aa-24aa-44e7-b485-81a988e7492f`
- ☒ Object ID: `f85873ef-78e3-4e6e-97dc-48c4f0e95139`
- ☒ Tenant ID: `c4a8886b-f140-478b-ac47-249555c30afd`

### Next Steps:

#### 1. Generate **Client Secret** (App Password)

- Go to Azure Portal → App registrations → ftc-ispilot-corp-prod
- Navigate to Certificates & secrets
- Create new client secret
- Copy immediately (cannot be retrieved later)
- This becomes `MICROSOFT_APP_PASSWORD`

#### 2. Create Teams Bot Channel

- Use App ID and Secret above
- Enable Teams channel
- Get messaging endpoint: `https://teams-ispilot-bridge-xxxxxx.run.app/api/messages`

See [NEXT\\_STEPS.md](#) → "Phase 1" for detailed checklist

## Local Development

### 1. Setup

```
cd teams_bot_bridge

# Create virtual environment
python -m venv .env
source .env/bin/activate # Windows: .env\Scripts\activate

# Install dependencies
pip install -r requirements.txt

# Copy environment template
cp .env.example .env
```

### 2. Configure Environment

Edit `.env`:

```
# Azure Bot Service credentials (from Phase 1)
MICROSOFT_APP_ID=<your-app-id>
MICROSOFT_APP_PASSWORD=<your-app-password>

# Google Cloud
GOOGLE_CLOUD_PROJECT=corp-stro-salesinventory-prod

# ispirot-api
ISPILOT_API_ENDPOINT=https://ispilot-api-46y2f3tyja-uc.a.run.app/chat

# Local dev
PORT=8080
DEBUG=True
```

### 3. Run Locally

```
uvicorn main:app --reload --host 0.0.0.0 --port 8080
```

Service will be available at: <http://localhost:8080>

### 4. Test Health Endpoint

```
curl http://localhost:8080/health
```

Expected response:

```
{
  "status": "ok"
}
```

## Cloud Run Deployment

### Deploy Command

```
PROJECT_ID="corp-stro-salesinventory-prod"
REGION="us-central1"
SERVICE_NAME="teams-ispilot-bridge"
SERVICE_ACCOUNT="sa-tot-osa@${PROJECT_ID}.iam.gserviceaccount.com"

gcloud run deploy "${SERVICE_NAME}" \
  --source . \
  --region "${REGION}" \
```

```
--project "${PROJECT_ID}" \  
--service-account "${SERVICE_ACCOUNT}" \  
--allow-unauthenticated \  
--memory 512Mi \  
--cpu 1 \  
--timeout 60 \  
--set-env-vars \  
  MICROSOFT_APP_ID=<APP_ID>,\  
  MICROSOFT_APP_PASSWORD=<APP_PASSWORD>,\  
  ISPILOT_API_ENDPOINT=https://ispilot-api-46y2f3tyja-uc.a.run.app/chat,\  
  GOOGLE_CLOUD_PROJECT=${PROJECT_ID}
```

## Get Service URL

```
gcloud run services describe teams-ispilot-bridge \  
  --region=us-central1 \  
  --project=corp-stro-salesinventory-prod \  
  --format='value(status.url)'
```

Use this URL as the **Messaging Endpoint** in Azure Bot Service registration.

## API Endpoints

### GET /health

Health check (no authentication).

```
curl https://teams-ispilot-bridge-xxxxxx.run.app/health
```

### POST /api/messages

Handles Teams activities from Azure Bot Service. **Requires valid JWT from Azure Bot Service.**

This is the endpoint configured in Azure Bot Service messaging configuration.

**Headers:** (Added by Azure Bot Service automatically)

```
Authorization: Bearer <JWT-from-Azure>  
Content-Type: application/json
```

**Request body:** (Activity JSON from Teams)

```
{  
  "type": "message",
```

```
"from": {"id": "user-id", "name": "User Name"},  
"conversation": {"id": "conversation-id"},  
"text": "How is Talca Colin performing?"  
}
```

**Response:** Activity response sent back to Teams.

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## Testing Sequence

### Step 1: Local Testing

1. Run locally: `uvicorn main:app --reload`
2. Verify health: `curl http://localhost:8080/health`
3. Check logs for startup messages

### Step 2: Cloud Run Testing

1. Deploy to Cloud Run (see deployment command above)
2. Get service URL: `gcloud run services describe teams-ispilot-bridge --region=us-central1 --format='value(status.url)'`
3. Test health: `curl https://<SERVICE-URL>/health`
4. Check logs: `gcloud run logs read teams-ispilot-bridge --region=us-central1 --limit 50`

### Step 3: Azure Bot Service Testing

1. Configure messaging endpoint in Azure Bot Service
2. Use Bot Framework Emulator to test
3. Or test in Teams Web Chat

### Step 4: Teams Testing

1. Publish Teams app (manifest.json configured)
2. Add app to team/group chat
3. Send test message
4. Verify response comes through

See [../docs/TEAMS\\_SDK\\_INTEGRATION.md](#) → "Testing Checklist" for complete validation steps.

## Troubleshooting

### Bridge not receiving messages

- **Check:** Azure Bot messaging endpoint is correctly configured
- **Fix:** Verify Cloud Run service URL in Azure Bot registration
- **Verify:** `gcloud run logs read teams-ispilot-bridge --region=us-central1`

### JWT validation fails

- **Check:** MICROSOFT\_APP\_ID and MICROSOFT\_APP\_PASSWORD are correct
- **Fix:** Regenerate credentials in Azure AD if unsure
- **Verify:** Check logs for JWT validation errors

## ispilot-api returns 401

- **Check:** sa-tot-osa has correct Workload Identity permissions
- **Fix:** Verify identity token generation works locally:

```
gcloud auth print-identity-token \
  --impersonate-service-account=sa-tot-osa@corp-stro-salesinventory-
  prod.iam.gserviceaccount.com
```

- **Verify:** ispilot-api README for IAM setup

## Message timeouts

- **Check:** ispilot-api is responding (see [../ispilot-api/README.md](#) smoke test)
- **Fix:** Increase Cloud Run timeout (currently 60s)
- **Verify:** Monitor ispilot-api logs:

```
gcloud run logs read ispilot-api --region=us-central1
```

## Environment Variables

| Variable               | Required                            | Description          | Default                                                                                                         |
|------------------------|-------------------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------|
| MICROSOFT_APP_ID       | <input checked="" type="checkbox"/> | Azure Bot App ID     | (none)                                                                                                          |
| MICROSOFT_APP_PASSWORD | <input checked="" type="checkbox"/> | Azure Bot App Secret | (none)                                                                                                          |
| ISPILOT_API_ENDPOINT   | <input checked="" type="checkbox"/> | ispilot-api URL      | <a href="https://ispilot-api-46y2f3tyja-uc.a.run.app/chat">https://ispilot-api-46y2f3tyja-uc.a.run.app/chat</a> |
| GOOGLE_CLOUD_PROJECT   | <input checked="" type="checkbox"/> | GCP Project ID       | corp-stro-salesinventory-prod                                                                                   |
| PORT                   |                                     | Server port          | 8080                                                                                                            |
| DEBUG                  |                                     | Enable debug logging | False                                                                                                           |

## References

- [Microsoft 365 Agents SDK Docs](#)
- [Azure Bot Service Docs](#)

- [Teams App Manifest Schema](#)
- [Google Cloud Identity Tokens](#)

## Implementation Status

See [../WORK\\_CHECKPOINT.md](#) for current implementation status and remaining tasks.